

Model Validation Interview Preparation

A Question List, with a Worked Discussion of the Kolmogorov–Smirnov Test

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Contents

1	About this document	2
2	Overview: MV Specializations	2
3	Linear Regression: Assumptions and Violations	2
3.1	The Classical Assumptions (Gauss-Markov)	2
3.2	Consequences of Violations	3
4	Logistic Regression: Assumptions	3
5	Hypothesis Testing: Core Concepts	4
5.1	Type I and Type II Errors	4
5.2	p-value	4
5.3	Statistical Power	4
6	Probability: Foundational Definitions	4
6.1	Marginal and Conditional Probability	4
6.2	Law of Total Probability	4
6.3	Independence	4
6.4	Integration Using the Kernel of a PDF	5
7	Structure of a Model Validation Project	5
8	Foundations of Statistical Testing	6
8.1	A Test Is a Bernoulli-Valued Statistic	6
8.2	Type I and II Errors; the Partial Order on Tests	6
8.3	Simple vs. Composite Hypotheses	6
8.4	The Role of Ω and the Sample Size n	7
8.5	p-value: Definition and Limitations	7
8.6	Consistency of a Test Sequence	8
8.7	Pivotal Quantities and the Traditional Test Statistic	8
9	The Kolmogorov-Smirnov Test	8
9.1	The Fundamental Fact: Distribution-Free Property	8
9.2	Two Convergence Results	9
9.3	The Test Statistic	10
9.4	Two Decision Regimes	10
9.5	Worked Example	11
9.6	Simulation Code (Python)	11
10	Quick Reference: Diagnostic Tests	12

1. About this document

A friend who works in model validation at a Canadian bank walked me through the question list he uses when interviewing MV candidates, and encouraged me to publish my notes from that conversation. Here they are.

The format is simple: I went down the list and wrote up the items that either interested me or that I felt I should know better. The Kolmogorov–Smirnov test (Section 9) got the most attention because my own understanding of it was the weakest going in. Other items are sketched lightly, sometimes only as a paragraph. Whether a topic gets one paragraph or six pages reflects where I happened to have a gap, not the topic’s importance.

What is most likely to come next is a fuller treatment of homoscedasticity — the assumption itself and the Breusch–Pagan test for it — which is what I am working on right now.

If you spot a mistake or want to add a discussion of your own, open an issue at <https://github.com/borisgarbuzov/notes> or comment on the LinkedIn post.

2. Overview: MV Specializations

Model validation (MV) roles vary significantly by the type of model being validated. The main domains and their primary modelling paradigms are:

Domain	Model type	Key techniques
Credit Risk	Regression / scoring	PD, LGD, EAD models; logistic regression
Stress Testing	Macro-conditional regression	OSFI/DFAST scenarios; satellite models
Fraud Detection	Classification	Logistic, random forest, XGBoost
AML	Unsupervised learning	Clustering, anomaly detection
Option Pricing / Market Risk	Stochastic processes	BSM, Monte Carlo, Greeks
Treasury / ALM	Interest rate models	Duration, convexity, gap analysis

Credit Risk and Stress Testing were flagged as the highest-priority domains for preparation.

3. Linear Regression: Assumptions and Violations

3.1 The Classical Assumptions (Gauss-Markov)

Let the model be $Y = X\beta + \varepsilon$.

- (a) **Linearity.** The relationship between Y and each regressor is linear. *Diagnostic:* scatter plots of Y vs. each X_j ; residual-vs-fitted plot.
- (b) **No multicollinearity.** The regressors are not perfectly (or near-perfectly) linearly dependent. *Diagnostic:* Variance Inflation Factor

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is the R^2 from regressing X_j on all other regressors. Rule of thumb: $\text{VIF} > 10$ signals a problem.

- (c) **No autocorrelation of residuals.** $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. *Diagnostic:* **Durbin-Watson** statistic

$$DW = \frac{\sum_{t=2}^n (\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2}{\sum_{t=1}^n \hat{\varepsilon}_t^2}.$$

$DW \approx 2$ indicates no autocorrelation; $DW < 2$ positive, $DW > 2$ negative autocorrelation.

(d) **Residuals are normally distributed with zero mean.** $\varepsilon \sim N(0, \sigma^2 I)$. *Diagnostics:*

- Zero-mean: one-sample t -test on residuals.
- Normality: **Kolmogorov-Smirnov (KS) test**; also Shapiro-Wilk, Q-Q plot.

(e) **Homoscedasticity.** $\text{Var}(\varepsilon_i) = \sigma^2$ (constant variance).

Theoretical basis (from first principles). Write the model as $y = \beta_0 + \beta_1 x + \varepsilon$ with $\mathbb{E}\varepsilon = 0$. Introduce an auxiliary model for the variance:

$$\varepsilon^2 = h(\alpha_0 + \alpha_1 x) + u, \quad \mathbb{E}u = 0, \quad \mathbb{E}(u|x) = 0 \text{ a.s.}$$

Then:

$$\text{Var}(\varepsilon|x) = \mathbb{E}(\varepsilon^2|x) = \mathbb{E}(h(\alpha_0 + \alpha_1 x) + u | x) = h(\alpha_0 + \alpha_1 x) + 0 = h(\alpha_0 + \alpha_1 x).$$

Homoscedasticity by definition means $\text{Var}(\varepsilon|x) = \text{const}$, i.e. $h(\alpha_0 + \alpha_1 x) = \text{const}$. A natural sufficient condition is $\alpha_1 = 0$. The **Breusch-Pagan test** tests exactly this:

$$H_0: \alpha_1 = 0.$$

Diagnostic procedure: regress $\hat{\varepsilon}^2$ on the regressors; the test statistic $nR^2 \sim \chi_k^2$ under H_0 .

3.2 Consequences of Violations

- (a) *Multicollinearity violated:* $\hat{\beta}$ remains unbiased but is **no longer consistent** in the sense that $\hat{\beta} \rightarrow \Sigma$ (the coefficient estimates are unstable and highly sensitive to small changes in data). Standard errors are inflated; interpretation of individual coefficients is meaningless.
- (b) *Autocorrelation or heteroscedasticity violated (assumptions c and e):* $\hat{\beta}$ is still unbiased, but $\widehat{\text{Var}}(\hat{\beta})$ is **incorrect**, invalidating all t - and F -tests. *Remedy:* use **Newey-West (HAC)** standard errors, which are robust to both autocorrelation and heteroscedasticity without re-specifying the model.

4. Logistic Regression: Assumptions

Logistic regression models the probability of a binary outcome:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-X\beta}} = \sigma(X\beta).$$

Key assumptions:

- (a) **Binary (or ordinal) outcome.** The dependent variable must be categorical.
- (b) **No multicollinearity** among regressors (same as linear regression; check VIF).
- (c) **Independence of observations.**
- (d) **Large sample size** (MLE relies on asymptotic theory).
- (e) **No extreme outliers** (influential observations distort MLE).

Note: logistic regression does *not* require normality of residuals or homoscedasticity.

5. Hypothesis Testing: Core Concepts

5.1 Type I and Type II Errors

	H_0 true	H_0 false
Reject H_0	Type I error (α)	Correct (Power)
Fail to reject H_0	Correct	Type II error (β)

- **Type I error** (false positive): reject H_0 when it is true. Probability = α (significance level).
- **Type II error** (false negative): fail to reject H_0 when it is false. Probability = β .

5.2 p-value

The **p-value** is the probability of observing a test statistic at least as extreme as the one computed, *under the assumption that H_0 is true*:

$$p = P(T \geq t_{\text{obs}} \mid H_0).$$

A small p -value (typically $< \alpha = 0.05$) is evidence against H_0 . The p -value is *not* the probability that H_0 is true.

5.3 Statistical Power

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_0 \text{ is false}).$$

Power increases with: larger sample size n , larger effect size, lower σ , higher α .

6. Probability: Foundational Definitions

6.1 Marginal and Conditional Probability

- **Marginal probability:** $P(A)$ — the probability of event A regardless of any other event.
- **Conditional probability:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

- **Multiplication rule:**

$$P(A \cap B) = P(A \mid B) P(B) = P(B \mid A) P(A).$$

6.2 Law of Total Probability

If $\{B_1, B_2, \dots, B_n\}$ is a partition of the sample space (mutually exclusive and exhaustive), then

$$P(A) = \sum_{i=1}^n P(A \mid B_i) P(B_i).$$

6.3 Independence

Events A and B are **independent** if and only if any of the following equivalent conditions holds:

$$P(A \mid B) = P(A), \quad P(B \mid A) = P(B), \quad P(A \cap B) = P(A) P(B).$$

6.4 Integration Using the Kernel of a PDF

A useful technique: identify the *kernel* of a known distribution inside an integral and use the fact that the PDF integrates to 1.

Example — Normal PDF.

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The *kernel* is $\exp(-(x - \mu)^2/(2\sigma^2))$.

Classic interview integral: Evaluate

$$I = \int_{-\infty}^{\infty} e^{-(\frac{x-\mu}{\sigma})^2} dx.$$

Solution. Write $u = (x - \mu)/\sigma$, $dx = \sigma du$:

$$I = \sigma \int_{-\infty}^{\infty} e^{-u^2} du = \sigma \sqrt{\pi}.$$

The last step uses the Gaussian integral $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$.

More generally, since $\int_{-\infty}^{\infty} f(x \mid \mu, \sigma^2) dx = 1$:

$$\int_{-\infty}^{\infty} e^{-(x-\mu)^2/(2\sigma^2)} dx = \sqrt{2\pi} \sigma.$$

7. Structure of a Model Validation Project

The typical deliverables of an MV engagement are:

- 1. Data replication.** Independently reproduce the training/calibration dataset from source systems. Verify row counts, distributions, and key statistics across time periods. *Example:* model developer reports 2025 Q1 portfolio = \$1.0M, 2025 Q2 = \$0.8M. Validator independently reconstructs both figures; unexplained gap of $-\$0.2M$ triggers investigation.
- 2. Model replication.** Re-implement the model independently (or re-run developer code) to reproduce reported outputs: coefficients, predictions, performance metrics. Discrepancies indicate either data or implementation errors.
- 3. Assumption checks.** Formally test all model assumptions as per the validation guideline (e.g. OSFI E-23, SR 11-7). For regression models: linearity, no multicollinearity, no autocorrelation, normality, homoscedasticity (see Section 2).
- 4. Backtesting.** Compare model predictions against actual realised outcomes on out-of-time or out-of-sample data. *Metrics:* Gini coefficient, AUC-ROC, KS statistic (for credit), RMSE/MAE (for regression), PSI for population stability.
- 5. Benchmarking.** Compare the validated model against one or more challenger models. *Process:* model developers typically present a set of candidate models (e.g. models a, b, c, d, e) and select a champion. The validator independently constructs a benchmark (possibly a subset or alternative ranking, e.g. b, d, g, e, c) to assess whether the champion selection is defensible.
- 6. Challenges / findings.** Document material findings, assign risk ratings (low / medium / high), and issue recommendations or conditions on model use.

8. Foundations of Statistical Testing

This section sets out the mathematical skeleton of hypothesis testing before applying it to specific tests (Breusch–Pagan, KS). The observations below revisit material from graduate-level theoretical statistics in language adapted to the validation context.

8.1 A Test Is a Bernoulli-Valued Statistic

Let $(\Omega, \mathcal{F}, \{P_\theta\}_{\theta \in \Theta})$ be a statistical model. A **test** is any measurable function

$$\phi : \mathcal{X} \rightarrow \{0, 1\},$$

where $\mathcal{X}$ is the sample space ($\phi = 1$ means reject H_0 , $\phi = 0$ means do not reject). This is the minimal definition: no further structure is required.

Remark on randomised tests. One sometimes allows $\phi : \mathcal{X} \rightarrow [0, 1]$, interpreted as the probability of rejection given the observed sample. This is needed to achieve an exact size α in discrete models (e.g. binomial). The Neyman–Pearson lemma in its full generality requires randomised tests. In practice they are rarely used.

Trivial tests. $\phi \equiv 0$ (never reject) has $\alpha = 0$, $\beta = 1$. $\phi \equiv 1$ (always reject) has $\alpha = 1$, $\beta = 0$. Both are formally valid tests. Their existence shows that the Bernoulli property alone says nothing useful: we need additional constraints.

8.2 Type I and II Errors; the Partial Order on Tests

For a test ϕ and hypotheses H_0, H_1 :

$$\alpha(\phi) = \sup_{\theta \in H_0} E_\theta[\phi(X)], \quad \beta(\phi) = \sup_{\theta \in H_1} E_\theta[1 - \phi(X)].$$

α is the **size** (worst-case Type I error rate), β is the worst-case Type II error rate, and $1 - \beta$ is the **power**.

Tests are partially ordered: ϕ_1 *dominates* ϕ_2 if $\alpha(\phi_1) \leq \alpha(\phi_2)$ and $\beta(\phi_1) \leq \beta(\phi_2)$, with at least one strict. Most pairs of tests are incomparable in this order — one has smaller α , the other smaller β .

Standard resolution (Neyman–Pearson paradigm). Fix a tolerated Type I error α_0 (typically 0.05) and minimise β within the class $\{\phi : \alpha(\phi) \leq \alpha_0\}$. This resolves incomparability by making α the primary constraint and β the objective. The asymmetry is conventional, not mathematical: H_0 plays the role of the status quo, and the industry assigns it 95% survival against a challenger H_1 that must overcome the 5% hurdle.

Alternative: Bayesian weighting. Assign prior probabilities π_0, π_1 to H_0, H_1 and costs c_{01}, c_{10} to each error type. Minimise $\pi_0 c_{01} \alpha + \pi_1 c_{10} \beta$. This gives a different optimal threshold and resolves incomparability differently. The Neyman–Pearson approach is the frequentist special case with $\pi_0 c_{01} = \pi_1 c_{10}$ (symmetric costs) and hard constraint replacing soft penalty.

8.3 Simple vs. Composite Hypotheses

A **simple hypothesis** is a single measure: $H_0 = \{P_0\}$. A **composite hypothesis** is a set of measures: $H_0 = \{P_\theta : \theta \in \Theta_0\}$.

For *two simple hypotheses*, the Neyman–Pearson lemma gives the optimal test: threshold the likelihood ratio $\Lambda(x) = \frac{dP_1}{dP_0}(x)$ at some c . It exists, is essentially unique, and achieves maximum power at level α .

For composite hypotheses, Λ is no longer well-defined (which P_1 do you use?). The best general outcome is a **uniformly most powerful (UMP)** test: one that maximises power against every $\theta \in H_1$ simultaneously at level α_0 . UMP tests exist only in special cases — primarily one-parameter exponential families with monotone likelihood ratio (one-sided testing). In most multiparameter or two-sided problems, UMP does not exist, and one resorts to:

- **UMPU**: uniformly most powerful among unbiased tests.
- **LRT**: likelihood ratio test — maximise $\sup_{H_1} L$ vs. $\sup_{H_0} L$; asymptotically optimal under regularity conditions.
- **Traditional statistics**: t , F , χ^2 , derived from pivotal quantities and validated by experience.

8.4 The Role of Ω and the Sample Size n

The probability space $(\Omega, \mathcal{F}, P)$ is abstract. In practice we observe $X : \Omega \rightarrow \mathcal{X}$, and the test lives on $\mathcal{X}$. The connection to Ω is implicit.

A subtlety often swept under the rug: for a fixed statistic T (say, the sample mean), each sample size n gives a *different* test ϕ_n . Properties such as consistency and power are properties of the *sequence* $\{\phi_n\}_{n \geq 1}$, not of a single test.

8.5 p-value: Definition and Limitations

The **p-value** is informally “the probability of observing an outcome at least as extreme as the one observed, under H_0 .” Formally:

$$p(x) = \sup_{\theta \in H_0} P_\theta(T(X) \geq T(x)),$$

where T is a scalar test statistic and “ $\geq$ ” encodes the ordering of outcomes by their evidence against H_0 .

This definition requires:

- A scalar statistic T with a known (or parameter-free) distribution under H_0 .
- A sensible ordering: large values of T are more adverse to H_0 . This ordering is a partial order on outcomes; without it, “at least as extreme” is undefined.
- The distribution of T under H_0 to be the same for all $\theta \in H_0$ (pivotal quantity), otherwise the sup may be a conservative bound only.

When p-value is well-defined and exact. The KS test is the cleanest example: $T = D_n$ has the same distribution G_n for *every* continuous F_0 at every finite n (distribution-free property). No sup is needed; the p-value is exact.

When p-value is approximate or ill-defined. For composite H_0 with nuisance parameters, $P_\theta(T \geq t)$ depends on θ , and one takes the sup, yielding a conservative p-value. In high-dimensional or nonparametric settings, there may be no natural scalar T , and the p-value loses a clean interpretation.

An observation worth flagging: “further from H_0 ” $\nRightarrow$ “closer to H_1 .” The natural metric on the parameter space may not align with the power direction. A test optimal for detecting deviations in all directions simultaneously (e.g. a two-sided t -test) is suboptimal against a one-sided alternative. Conversely, a one-sided test has no power against the opposite direction. This is not a defect but a fundamental feature: without specifying H_1 , there is no universally best direction of extremeness.

8.6 Consistency of a Test Sequence

A sequence of tests $\{\phi_n\}$ at fixed level α is **consistent** against H_1 if the power $1 - \beta_n \rightarrow 1$ as $n \rightarrow \infty$ for every $\theta \in H_1$.

Clarification on the joint behaviour of α and β . Simultaneously driving both $\alpha_n \rightarrow 0$ and $\beta_n \rightarrow 0$ is possible if one relaxes the fixed-level constraint. For example, setting $\alpha_n = n^{-1/2}$ and raising the threshold accordingly can yield $\beta_n \rightarrow 0$ as well. But this is non-standard. The classical framework fixes α and asks only that $\beta_n \rightarrow 0$: consistency is a one-sided guarantee.

The reason for the asymmetry mirrors the earlier one: α is a hard constraint (we do not want to exceed the nominal false positive rate), while β is an objective (we want power, and we accept that more data is needed for harder alternatives).

8.7 Pivotal Quantities and the Traditional Test Statistic

A **pivotal quantity** is a function of the data and parameter whose distribution under H_0 does not depend on any unknown parameter. Examples: $\frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$ (Student's t , exact); $\sqrt{n} D_n \rightarrow K$ (KS, asymptotically pivotal).

When a pivot exists, it is natural to use it as the test statistic, because:

- Its null distribution is known — no simulation or bootstrap needed.
- The p-value is exact (or asymptotically exact).
- Critical values can be tabulated once.

The Bernoulli test statistic $\phi = \mathbf{1}[T > c_\alpha]$ is the final step. Tradition calls T “the test statistic” even though ϕ is the actual test. When no pivot exists, one still gets a Bernoulli test (e.g. via simulation), but the intermediate scalar T does not have the same clean properties.

9. The Kolmogorov-Smirnov Test

9.1 The Fundamental Fact: Distribution-Free Property

(This characterisation of the KS test as exactly distribution-free at every finite n comes from re-reading Kolmogorov's original 1933/1941 paper in the course of preparing these notes; it is rarely emphasised in textbooks, which usually present only the asymptotic Kolmogorov-distribution result.)

The entire KS test rests on one remarkable observation.

Theorem (Kolmogorov 1933). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$ where F is *any* continuous CDF. Then the distribution of $D_n = \sup_x |F_n(x) - F(x)|$ does **not depend on F** .

Proof sketch. The probability integral transform gives $U_i = F(X_i) \sim \text{Uniform}(0, 1)$. Applying F to every observation maps the original sample to a uniform sample, and D_n computed against F equals D_n computed for the uniform sample against $\text{Uniform}(0, 1)$. The distribution of the latter depends only on n .

Why this is stronger than the asymptotic statement. The theorem says that for *every fixed* n , the distribution G_n of D_n is the same for all continuous F . This is not just a statement about the limit $n \rightarrow \infty$: it holds exactly at $n = 5$, $n = 10$, $n = 100$, or any other sample size.

Consequences.

- **Exact tabulation for any n .** Because G_n is universal, it can be computed once and tabulated. This is the origin of the exact critical values $D_{n,\alpha}$ (double-indexed by n and significance level α):

$$P_{H_0}(D_n > D_{n,\alpha}) = \alpha,$$

the same equation for every continuous F_0 .

- **Asymptotic regime.** As $n \rightarrow \infty$, G_n converges to the Kolmogorov distribution G (Theorem 2 below), giving the large-sample formula.
- **Discontinuous F_0 .** If F_0 is not continuous, D_n depends on F_0 . However, Kolmogorov (1941, Theorem 3) showed $P(D_n \leq \lambda) \geq G_n(\lambda)$ for all λ , so the test remains conservative.

A note on the source. “Confidence Limits for an Unknown Distribution Function,” *Annals of Math. Stat.*, 1941, is three pages without proofs — a short communication summarising Kolmogorov’s 1933 Italian paper (*Giornale dell’Istituto Italiano degli Attuari*). Theorem 2 states convergence “uniformly in λ ” with no domain restriction; as our partial-sum analysis shows, convergence holds for $\lambda > 0$ only (at $\lambda = 0$ the partial sums oscillate between 0 and 2). Read it as an authoritative reference card, not a self-contained proof.

9.2 Two Convergence Results

Theorem 1 (Glivenko–Cantelli).

$$\sup_x |F_n(x) - F(x)| \xrightarrow{a.s.} 0.$$

Pointwise convergence follows from the LLN; the theorem strengthens it to *uniform* convergence: for almost all ω , the whole staircase F_n converges to F as a function.

Theorem 2 (Kolmogorov–Smirnov).

$$\sqrt{n} D_n \xrightarrow{d} K, \quad K = \sup_{t \in [0,1]} |B_t|, \quad B_t = W_t - tW_1 \text{ (Brownian bridge)}.$$

The factor $\sqrt{n}$ compensates the $O(1/\sqrt{n})$ rate from Theorem 1. The limiting distribution K is the **Kolmogorov distribution**: parameter-free, defined via the Brownian bridge.

Survival function of K . Two complementary series (both valid for $t > 0$, related by the Jacobi theta identity):

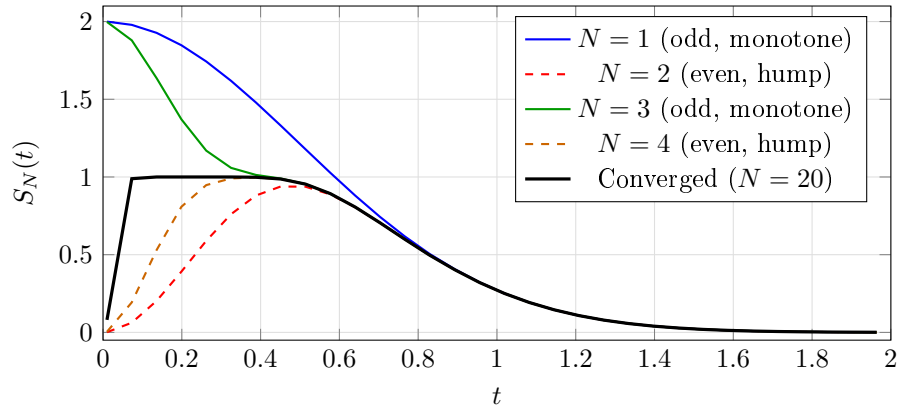
$$P(K \geq t) = 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 t^2} \quad (\text{fast for large } t)$$

$$P(K \leq x) = \frac{\sqrt{2\pi}}{x} \sum_{k=1}^{\infty} e^{-(2k-1)^2 \pi^2 / (8x^2)} \quad (\text{fast for small } x)$$

For $t \geq 1.2$ the first term of Series A alone is accurate to 10^{-4} .

Partial sums and monotonicity. Individual partial sums of Series A are not survival functions. Odd- N truncations ($N = 1, 3, 5, \dots$) start near 2 near the origin and decrease monotonically; even- N truncations ($N = 2, 4, 6, \dots$) start near 0, rise to a hump, then decrease. As $N \rightarrow \infty$ the hump migrates toward $t = 0$; convergence is on $(0, \infty)$ only (at $t = 0$ the series diverges).

Partial sums: $S_N(t) = 2 \sum_{k=1}^N (-1)^{k-1} e^{-2k^2 t^2}$

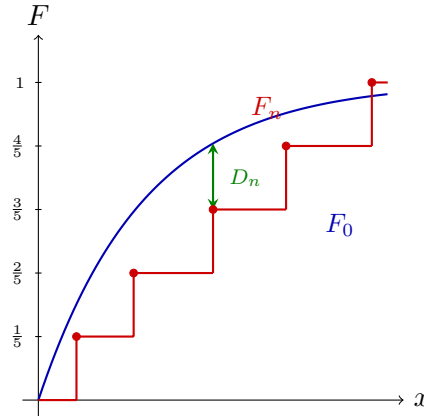


9.3 The Test Statistic

Setting. $H_0: F = F_0$ where F_0 is fully specified and continuous.

$$D_n = \sup_x |F_n(x) - F_0(x)| = \max_{1 \leq i \leq n} \max\left\{\left|\frac{i}{n} - F_0(x_{(i)})\right|, \left|\frac{i-1}{n} - F_0(x_{(i)})\right|\right\}.$$

Note: $|F_n - F_0| = |(1 - F_n) - (1 - F_0)|$, so comparing CDFs is identical to comparing survival functions.



Staircase F_n (red) vs. F_0 (blue). Arrow marks D_n .

9.4 Two Decision Regimes

Small samples ($n \lesssim 40$) — **exact table.** Look up $D_{n,\alpha}$ and reject H_0 if $D_n > D_{n,\alpha}$. No $\sqrt{n}$ scaling. The table exists because G_n is universal.

n	$\alpha = 0.20$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
5	0.447	0.509	0.565	0.669
10	0.323	0.369	0.409	0.486
20	0.232	0.265	0.294	0.352
40	0.165	0.189	0.210	0.252

Large samples ($n \gtrsim 40$) — **Kolmogorov asymptotics.** Form $T = \sqrt{n} D_n$ and reject if $T > c_\alpha$, or compute $p = P(K \geq T) = 2 \sum_{k=1}^{\infty} (-1)^{k-1} e^{-2k^2 T^2}$ (one term suffices for $T \geq 1.2$).

α	c_α	Reject if $\sqrt{n} D_n >$
0.10	1.224	1.224
0.05	1.358	1.358
0.01	1.628	1.628

9.5 Worked Example

Setup. $H_0: F_0 = \text{Exp}(1)$, so $F_0(x) = 1 - e^{-x}$.

Path A: $n = 10$ (exact table)

Sorted mock data: 0.11, 0.28, 0.43, 0.57, 0.74, 0.95, 1.21, 1.58, 2.03, 2.87.

i	$x_{(i)}$	$F_n = i/10$	$F_0(x_{(i)})$	$ F_n - F_0 $
1	0.11	0.10	0.104	0.004
2	0.28	0.20	0.244	0.044
3	0.43	0.30	0.349	0.051
4	0.57	0.40	0.435	0.035
5	0.74	0.50	0.523	0.023
6	0.95	0.60	0.613	0.013
7	1.21	0.70	0.702	0.002
8	1.58	0.80	0.794	0.006
9	2.03	0.90	0.869	0.031
10	2.87	1.00	0.943	0.057

$D_{10} = 0.057$. From exact table: $D_{10,0.05} = 0.409$. Since $0.057 < 0.409$: **fail to reject** H_0 .

Path B: $n = 100$ (Kolmogorov asymptotics)

Mock: $D_{100} = 0.121$. Scaled: $T = \sqrt{100} \cdot 0.121 = 1.21$. Since $T \geq 1.2$, one term: $p \approx 2e^{-2(1.21)^2} = 0.107$. Since $T = 1.21 < c_{0.05} = 1.358$: **fail to reject** H_0 . If instead $D_{100} = 0.148$, then $T = 1.48 > 1.358$: **reject**.

9.6 Simulation Code (Python)

```
def empirical_cdf(sample):
    n = len(sample)
    @np.vectorize
    def Fn(x): return (1/n) * np.sum(sample <= x)
    return Fn

def KS_stat(sample, F):
    # returns sqrt(n)*D_n
    n = len(sample); Fn = empirical_cdf(sample)
    grid = np.arange(np.min(sample)-1, np.max(sample)+1, 0.01)
    return np.sqrt(n) * np.max(np.abs(Fn(grid) - F(grid)))

def Brownian_bridge(t):
    # B_t = W_t - t*W_1
    W = np.cumsum(np.sqrt(np.diff(t)) * nr.normal(0,1,len(t)-1))
    W = np.r_[0, W]
    return W - W[-1] * t

def KS_rv():
    # one draw of K = sup|B_t|
```

```

return np.max(np.abs(Brownian_bridge(np.arange(0, 1, 1e-5))))

def KS_cdf():
    # empirical KS CDF from 10,000 paths
    return empirical_cdf(np.array([KS_rv() for _ in range(10000)]))

```

The simulation confirms the distribution-free property in its full strength: for every fixed n , the distribution of D_n is *exactly the same* regardless of whether the data are Normal, Exponential, or any other continuous distribution. The Kolmogorov distribution G is only the *limit* of G_n as $n \rightarrow \infty$; the universality of G_n at each finite n is the deeper fact, and it is what makes the exact small-sample tables possible.

The full runnable notebook — with the same functions extended for typing and reproducibility, plots of the empirical and theoretical CDFs, the simulated KS distribution, and convergence diagnostics at $n = 10, 100, 1000$ — is at https://github.com/borisgarbuzov/notes/blob/main/KS_test.ipynb.

10. Quick Reference: Diagnostic Tests

Assumption	Test / Method	What to check
Linearity	Scatter plot, residual vs. fitted	Visual pattern
Multicollinearity	VIF	$VIF > 10$
Autocorrelation	Durbin-Watson	$DW \approx 2$
Zero mean of residuals	One-sample t -test	$p > 0.05$
Normality of residuals	KS test, Shapiro-Wilk, Q-Q	$p > 0.05$
Homoscedasticity	Breusch-Pagan	$p > 0.05$
Population stability	PSI	PSI < 0.1 stable, > 0.25 unstable
Discrimination (credit)	Gini / AUC, KS stat	Gini > 0.3 acceptable